

# Drawing the Last Straw: Competition and Deception in a Duopoly

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Deception, such as hidden fees or misleading product names, makes an offer look better, but only up to a point. The deception that pushes a consumer past it is the last straw, and she abandons the firm. How much it takes depends on what she knows, and firms rarely know how much their customers will notice. We study a duopoly in which consumers buy from the most deceptive firm whose deception they have not detected. When firms know how discerning consumers are, competition escalates deception, and in the mixed-strategy equilibrium average deception exceeds the monopoly level by about 41 per cent. When they do not, uncertainty cuts both ways. If an undiscerning market is unlikely, firms deceive less on average, and earn less, than if they knew. If it is likely enough, firms gamble on it, and consumers periodically see through a firm and abandon it outright. These collapses are an equilibrium outcome, and they are most likely when firms are fairly, but not fully, confident that their customers will not notice.

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## 1. Introduction

Much of what firms do to sell a product leaves the product itself unchanged. A ticketing site adds its fees at checkout; a shopping site runs a countdown timer that resets whenever the page is reloaded; a subscription takes one click to start and a string of screens to cancel. A crawl of roughly 11,000 shopping websites found such manipulative designs, known as dark patterns, on about one in nine of them ([Mathur et al., 2019](#)). Regulators have begun to act, and in 2025 Amazon agreed to pay \$2.5 billion to settle U.S. Federal Trade Commission charges that it had enrolled consumers in Prime without their consent and made it hard to cancel ([Federal Trade Commission, 2025](#)).

Three features of such deception motivate this paper. First, it works only for as long as it goes unnoticed. In a large online experiment, mild dark patterns more than doubled the share of subjects who signed up for a dubious data-protection plan, while aggressive ones, which subjects were more likely to recognize for what they were, provoked a strong backlash ([Luguri and Strahilevitz, 2021](#)). Second, how much deception goes unnoticed depends on what a consumer knows. Processor makers name their chips so that an older, slower part can sound like a newer, faster one. To many buyers, “i7” simply means a good processor, whatever its generation, so a misleading name slips past them easily, but a buyer who follows the hardware sees through the same name at once. Third, a firm cannot tell in advance how discerning its own customers are, so every new fee, design or product name is in part a bet on how much they will notice.

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This paper asks how competition shapes deception when consumers behave this way, and what changes when firms do not know how much deception their consumers will detect. Two symmetric firms simultaneously choose a level of deception, meaning any practice that raises a product's perceived attractiveness without changing the product. Each consumer has a private tolerance threshold, the most deception she can be subjected to without noticing it, and thresholds are uniformly distributed up to a ceiling  $\theta$ . The less a consumer knows, the higher her threshold. Following [Kahneman et al. \(1997\)](#), we separate the decision utility on which a consumer chooses from the experienced utility she eventually receives. Below her threshold, deception raises the first while lowering the second, so she sees a more attractive offer even though she ends up worse off. Above her threshold, she detects the deception and rejects the firm outright. She therefore buys from the most deceptive firm whose deception she has not detected. Deception thus raises the revenue extracted from each customer, drives away the marginal customer whose threshold it crosses, and, under competition only, lets a firm overtake its rival and serve every consumer who does not detect the higher level.

*Complete information.* When firms know  $\theta$ , the overtaking motive rules out equilibrium in pure strategies (Proposition 1). A firm that matches its rival's level shares the top of the market, which it could capture whole by deceiving slightly more, and the less deceptive of two firms gains by leapfrogging. Firms therefore randomize, much as sellers randomize prices in [Varian \(1980\)](#), except that customers here flock to the firm that overtakes rather than the one that undercuts. We characterize the symmetric mixed-strategy equilibrium and show that competition escalates deception rather than disciplining it, contrary to the common presumption that a less deceptive rival keeps firms honest. With linear revenue, average deception exceeds the monopoly level by about 41 per cent and the highest level played exceeds it by about 71 per cent (Corollary 1). The escalation is largest when revenue responds least to deception, since a monopolist then barely deceives while duopolists keep deceiving to overtake each other.

*Incomplete information.* The main focus of the paper is the case in which firms do not observe  $\theta$  (Section 5). The market is either fragile, with ceiling  $\theta_F$ , or robust, with a higher ceiling  $\theta_R$ . Consumers in a fragile market see through deception sooner, for instance because more of them know the product well. Both firms know only the probability  $\mu$  that the market is robust. A firm that deceives beyond  $\theta_F$  is therefore gambling. If the market turns out to be fragile, every consumer sees through it and walks away. Because demand is linear in the distribution of tolerance, the uncertain game is equivalent to a known market whose tolerance distribution is the average of the two (Lemma 1), which lets us solve it with the tools of the complete information case.

When a robust market is unlikely, firms stay below  $\theta_F$  and behave as if the ceiling were the harmonic mean of  $\theta_F$  and  $\theta_R$  (Proposition 4). Uncertainty averages the rates at which deception loses customers, not the ceilings themselves, and this pulls firms towards caution. They deceive less on average, and earn less than if they knew the market (Proposition 5). When a robust market is likely enough, caution gives way to gambling. Firms deceive past  $\theta_F$  with positive probability, and whenever the market turns out to be fragile, an overreaching firm loses its entire market, an event we call a bust. Busts are an equilibrium outcome, not a mistake, and their probability is hump-shaped in firms' confidence, reaching 58 per cent in one of our examples (Figure 2b). When consumers in the robust market are much less discerning, each firm's strategy also splits in two. It either stays safely below  $\theta_F$  or deceives far beyond it, because once the fragile market's customers are lost, further deception becomes cheap (Figure 2a).

These results bear on policy in three ways. Competition is no substitute for consumer protection, since adding a rival raises deception rather than lowering it, and consumers' own choices will not correct this, since they respond to the decision utility that deception inflates. Where firms are cautious, tools that let them learn how much their customers notice, such as testing interface designs on live traffic, push deception up. And episodes in which consumers abandon a firm need not mean the firm misjudged them. Such episodes are most frequent when firms are fairly, but not fully, confident that consumers will not notice more.

The paper belongs to behavioral industrial organization, where a recurring theme is that competition need not eliminate deception. In [Gabaix and Laibson \(2006\)](#), firms keep add-on prices shrouded because educating consumers does not pay, and [Akerlof and Shiller \(2015\)](#) argue more broadly that competition ensures every profitable opportunity to manipulate consumers is taken. We find that competition does more than sustain deception, since it raises deception above the monopoly level. The closest setting is [Glaeser and Ujhelyi \(2010\)](#), in which firms misinform consumers and every consumer buys from the firm that has misled her most; they note that simultaneous choices then admit no pure-strategy equilibrium and turn to a sequential game. Because our consumers have limits, the less deceptive firm keeps the customers its rival drives away, which lets us solve the simultaneous game directly, under both complete and incomplete information.

## 2. The Model

We consider a static duopoly game of incomplete information. Two symmetric firms, indexed by  $i \in \{1, 2\}$ , compete by simultaneously choosing a level of strategic deception  $d_i \in [0, \bar{d}]$ , where  $\bar{d}$  represents the structural upper bound of feasible deceptive practices.

### 2.1. Consumer Behavior and Demand Microfoundations

The market consists of a continuum of heterogeneous consumers normalized to a total mass of one. 'Deception' in this market is defined as an action taken by a firm that increases the perceived attractiveness of a product to consumers without intrinsically changing the product itself. Each consumer is characterized by a private tolerance threshold  $\tau$ , which represents the maximum level of deception they will tolerate before completely rejecting a firm's product. We assume that consumer tolerance is uniformly distributed across the population up to an aggregate market limit  $\theta$ , such that  $\tau \sim U[0, \theta]$ .

A consumer facing a firm playing deception level  $d_i$  evaluates their options according to their decision utility:

**Assumption 1** (Consumer Decision Utility). The utility a consumer with private tolerance  $\tau$  derives from purchasing from firm  $i$  is given by

$$\tilde{U}(d_i, \tau) = \begin{cases} v(d_i) & \text{if } d_i \leq \tau \\ 0 & \text{if } d_i > \tau \end{cases} \quad (1)$$

with  $v'(d_i) > 0$ . Since we only model consumer choice, experienced utility is not modeled directly. However, we assume that it is a decreasing function in deception, i.e. deception inflates the perceived value of the product when making the choice to buy, but reduces actual experienced utility once the consumer eventually realizes they've been deceived.

A higher level of  $d_i$  artificially inflates the product's perceived attractiveness, driving up the consumer's perceived utility. This psychological illusion persists continuously until the firm overreaches and crosses the consumer's private tolerance threshold  $\tau$ . At this point, the consumer detects the deception and ceases to buy from the firm.

Let  $d_H = \max\{d_1, d_2\}$  and  $d_L = \min\{d_1, d_2\}$  denote the high and low deception firms. Given that consumers maximize utility subject to their tolerance constraint, the aggregate demand schedules  $q_H$  and  $q_L$  are determined by the realization of  $\theta$ .

$$q_H(d_H; \theta) = \begin{cases} \frac{\theta - d_H}{\theta}, & d_H \leq \theta, \\ 0, & d_H > \theta, \end{cases} \quad q_L(d_L, d_H; \theta) = \begin{cases} \frac{d_H - d_L}{\theta}, & d_H \leq \theta, \\ \frac{\theta - d_L}{\theta}, & d_L \leq \theta < d_H, \\ 0, & d_L > \theta. \end{cases} \quad (2)$$

## 2.2. Firm Payoffs and the Optimization Problem

Firms are risk-neutral and maximize expected profits. Deception operates as a double-edged sword: it reduces consumer demand, but increases the revenue extracted per retained customer. We formalize this revenue a firm can extract *because* deception inflates perceived value as  $r(d_i) = d_i^\gamma$ , where  $\gamma > 0$ . We will make two useful assumptions on  $r$ :

- (i)  $r(0) = 0$ , and  $r$  is positive, differentiable and strictly increasing on  $(0, \bar{d}]$ ,
- (ii)  $\frac{r(d)^2}{r'(d)}$  is strictly increasing on  $(0, \bar{d}]$ .

Firms face no production costs. The profit function for firm  $i$  is given by:

$$\pi_i(d_i, d_j; \theta) = r(d_i) \cdot q_i(d_i, d_j; \theta) \quad (3)$$

## 3. Solving the model with complete information

### 3.1. Monopoly benchmark

A single firm in this environment would maximize  $\pi = d^\gamma \frac{\theta - d}{\theta}$ . Maximizing this, we get the result

$$d^{mon} = \frac{\gamma}{\gamma + 1} \theta, \quad \pi^{mon} = \frac{\gamma^\gamma}{(\gamma + 1)^{\gamma+1}} \theta^\gamma$$

### 3.2. Pure Strategy Nash Equilibrium

**Proposition 1.** *For any strictly increasing continuous  $r(d)$ , the game has no pure strategy Nash equilibrium.*

*Proof.* If  $d_1 = d_2 = d$  and  $d < \theta$ , both firms share the consumers tolerant to  $d$  equally, each earning  $\pi_i = r(d) \frac{\theta - d}{2\theta}$ . Consider a deviation to  $d + \varepsilon$ . Profit for the deviator now becomes  $\pi_i = r(d + \varepsilon) \frac{\theta - d - \varepsilon}{\theta}$  as they no longer have to share the market. As  $\varepsilon \rightarrow 0$ ,  $\pi_i = r(d) \frac{\theta - d}{\theta}$ , which is twice as high. Therefore, a small positive deviation is strictly profitable.

If  $d_L < d_H$  and  $d_H > \theta$ , the high firm earns zero and can deviate to below  $\theta$  to earn positive profits. If  $d_H < \theta$ , we know that at equilibrium,  $\pi_H(d_H) > \pi_H(d) \forall d$ . In particular,  $\pi_H(d_H) >$

$\pi_H(d_L)$  for the high firm. Now for the low firm, it earns  $\pi_L = r(d_L) \frac{d_H - d_L}{\theta}$ . But notice this is strictly less than the numerator of  $\pi_H(d_L) = r(d_L) \frac{\theta - d_L}{\theta}$  as  $d_H < \theta$ . Therefore, we get  $\pi_H(d_H) > \pi_H(d_L) > \pi_L(d_L)$ . If the low firm plays  $d_H + \varepsilon$ , it becomes the new high firm and earns a higher profit than being the low firm, giving it an incentive to deviate.  $\square$

## 4. Mixed Strategy Nash Equilibrium

### 4.1. Indifference condition

Suppose the firm earns the same  $\pi^*$  across the entire support  $[\underline{s}, \bar{s}]$ . If  $F(d)$  denotes the CDF from which the firms draw actions, then their expected profit becomes

$$\mathbb{E}(\pi) = r(d) \left[ F(d) \frac{\theta - d}{\theta} + (1 - F(d)) \mathbb{E} \left[ \frac{x - d}{\theta} \mid x > d \right] \right] = \pi^* \quad \forall d \in [\underline{s}, \bar{s}] \quad (4)$$

To solve for MSNE, we need to set  $\mathbb{E}'(\pi) = 0$ . We'll do so by differentiating elements of  $\mathbb{E}(\pi)$  one by one. If  $f(d)$  is the density of  $F(d)$ , we can rewrite the conditional expectation. Let's call it the expected residual demand, represented by  $R(d)$ .

$$(1 - F(d)) \mathbb{E} \left[ \frac{x - d}{\theta} \mid x > d \right] = \int_d^{\bar{s}} \frac{x - d}{\theta} f(x) dx \equiv R(d)$$

### 4.2. Solving for the density

Differentiating it, we get

$$\begin{aligned} \frac{d}{dd} \int_d^{\bar{s}} \frac{x - d}{\theta} f(x) dx &= \frac{1}{\theta} \left[ -(d - d)f(d) + \int_d^{\bar{s}} \frac{\partial}{\partial d} [(x - d)f(x)] dx \right] \\ &= \frac{1}{\theta} \left[ \int_d^{\bar{s}} -(f(x)) dx \right] \\ &= \frac{1}{\theta} [-(F(\bar{s}) - F(d))] \\ \implies R'(d) &= \frac{-(1 - F(d))}{\theta} \end{aligned}$$

Next, differentiating  $F(d) \frac{\theta - d}{\theta} + R(d)$  gives us

$$\begin{aligned} \frac{d}{dd} \left[ F(d) \frac{\theta - d}{\theta} + R(d) \right] &= f(d) \frac{\theta - d}{\theta} - \frac{F(d)}{\theta} + R'(d) \\ &= f(d) \frac{\theta - d}{\theta} - \frac{F(d)}{\theta} - \frac{1 - F(d)}{\theta} \\ &= \frac{f(d)(\theta - d) - 1}{\theta} \end{aligned} \quad (5)$$

Finally, differentiating  $\mathbb{E}(\pi)$  yields

$$\mathbb{E}'(\pi) = r'(d) \left[ F(d) \frac{\theta - d}{\theta} + R(d) \right] + r(d) \left[ \frac{f(d)(\theta - d) - 1}{\theta} \right] = 0$$

Since  $\left[F(d)\frac{\theta-d}{\theta} + R(d)\right] = \frac{\pi^*}{r(d)}$ , we can clean up this expression to get

$$\begin{aligned}\mathbb{E}'(\pi) &= \frac{r'(d)\pi^*}{r(d)} + \frac{r(d)}{\theta}[f(d)(\theta-d) - 1] = 0 \\ \implies &\frac{r'(d)\pi^*\theta}{r(d)^2} + f(d)(\theta-d) - 1 = 0\end{aligned}$$

Therefore, we get the density

$$f(d) = \frac{1}{\theta-d} \left[ 1 - \frac{r'(d)\pi^*\theta}{r(d)^2} \right] \quad (6)$$

### 4.3. Identifying the optimal profit

Our next goal will be to pin down  $\pi^*$ . To do so, we can evaluate it at  $F(\underline{s})$  and  $F(\bar{s})$ , values of  $F$  we know. At  $d = \underline{s}$ ,  $F(\underline{s}) = 0$ , giving us

$$\begin{aligned}\pi^* &= r(\underline{s}) \int_{\underline{s}}^{\bar{s}} \frac{x-\underline{s}}{\theta} f(x) dx \\ &= \frac{r(\underline{s})}{\theta} \left[ \int_{\underline{s}}^{\bar{s}} x f(x) dx - \underline{s} \int_{\underline{s}}^{\bar{s}} f(x) dx \right] \\ &= \frac{r(\underline{s}) [\mathbb{E}[d] - \underline{s}]}{\theta}\end{aligned}$$

For any  $d < \underline{s}$ , we know that  $F(d) = 0$ , so we get a payoff of the same form  $\pi(d) = \frac{r(d) [\mathbb{E}[d] - d]}{\theta}$ . Since this  $d$  is below the support, it is not in the optimal interval for the MSNE, and thus it must earn a strictly lower profit than  $\pi(\underline{s}) = \pi^*$ . In other words,  $\underline{s}$  maximizes profits, and thus also maximizes  $\frac{r(d) [\mathbb{E}[d] - d]}{\theta}$  for  $d \leq \underline{s}$ . Differentiating, the FOC is  $r'(\underline{s}) [\mathbb{E}(d) - \underline{s}] - r(\underline{s}) \geq 0$  at the right side boundary, and thus we get

$$\mathbb{E}[d] - \underline{s} \geq \frac{r(\underline{s})}{r'(\underline{s})}$$

Substituting into the expression for  $\pi^*$ , we get

$$\pi^* \geq \frac{r^2(\underline{s})}{r'(\underline{s})\theta}$$

But from the density (Equation 6),  $f(\underline{s}) \geq 0 \implies \frac{r^2(\underline{s})}{r'(\underline{s})\theta} \geq \pi^*$ . Combining the two inequalities, we get

$$\pi^* = \frac{r^2(\underline{s})}{r'(\underline{s})\theta}$$

At  $d = F(\bar{s}) = 1$ ,  $\pi^* = r(\bar{s})\frac{\theta - \bar{s}}{\theta}$ , so

$$\pi^* = \frac{r(\bar{s})(\theta - \bar{s})}{\theta}$$

Since  $\pi^* > 0$ , this establishes that the upper end of the support  $\bar{s} < \theta$ , so firms never deceive past the market boundary in a complete information MSNE.

**Proposition 2** (Expected value of deception). *Under complete information,  $\mathbb{E}[d] = \frac{\gamma+1}{\gamma} \underline{s}$ .*

*Proof.* We know

$$\pi^* = \frac{r(\underline{s})[\mathbb{E}[d] - \underline{s}]}{\theta} = \frac{r^2(\underline{s})}{r'(\underline{s})\theta}.$$

This gives us

$$\mathbb{E}[d] = \underline{s} + \frac{r(\underline{s})}{r'(\underline{s})} = \frac{\gamma+1}{\gamma} \underline{s}. \quad \square$$

#### 4.4. With constant elasticity revenue

To move further, let's use our assumed functional form  $r(d) = d^\gamma$ . This gives us

$$\pi^* = \frac{\underline{s}^{\gamma+1}}{\gamma\theta} = \frac{\bar{s}^\gamma(\theta - \bar{s})}{\theta}$$

and our density is

$$f(d) = \frac{1}{\theta - d} \left[ 1 - \left( \frac{\underline{s}}{d} \right)^{\gamma+1} \right]$$

Integrating over the whole support gives us

$$\int_{\underline{s}}^{\bar{s}} \frac{1}{\theta - d} \left[ 1 - \left( \frac{\underline{s}}{d} \right)^{\gamma+1} \right] dd = 1$$

For simplicity, let  $\gamma = 1$ . Then,  $f(d) = \frac{1}{\theta - d} - \frac{\underline{s}^2}{d^2(\theta - d)}$ . Integrating this gives us

$$F(d) = \left( \left( \frac{\underline{s}}{\theta} \right)^2 - 1 \right) \ln(\theta - d) - \left( \frac{\underline{s}}{\theta} \right)^2 \ln(d) + \left( \frac{\underline{s}}{\theta} \right)^2 \frac{\theta}{d} + C$$

Since we know  $F(\bar{s}) - F(\underline{s}) = 1$ , subtracting gives us

$$\left( \rho^2 - 1 \right) \ln \left( \frac{\theta - \bar{s}}{\theta - \underline{s}} \right) - \rho^2 \ln \left( \frac{\bar{s}}{\underline{s}} \right) + \rho^2 \theta \left( \frac{1}{\bar{s}} - \frac{1}{\underline{s}} \right) = 1$$

where  $\rho \equiv \frac{\underline{s}}{\theta}$ . From  $\pi^*$ , we get  $\underline{s}^{\gamma+1} = \gamma \bar{s}^\gamma (\theta - \bar{s})$ . Therefore, the above expression is an equation in one variable which can be solved numerically. Doing so for  $\theta = 1$ , we get  $\bar{s} = 0.8557$  and thus  $\underline{s} = 0.3514$ .

**Proposition 3** (Scale invariance). *Fix  $\gamma > 0$  and let  $\theta > 0$  be arbitrary. If  $F_1$  is an equilibrium distribution for a market with tolerance ceiling 1, then the distribution  $F_\theta$  defined by  $F_\theta(d) = F_1(d/\theta)$  is an equilibrium distribution for a market with ceiling  $\theta$ , and conversely. Consequently*

$$\underline{s}(\theta) = \theta \underline{s}(1), \quad \bar{s}(\theta) = \theta \bar{s}(1), \quad \mathbb{E}_\theta[d] = \theta \mathbb{E}_1[d], \quad \pi^*(\theta) = \theta^\gamma \pi^*(1).$$

*Proof.* Write actions as  $d_i = \theta u_i$  with  $u_i \in [0, 1]$ . Since the demand schedules depend on deception only through its ratio to the ceiling,

$$q_H = \frac{\theta - d_H}{\theta} = 1 - u_H, \quad q_L = \frac{d_H - d_L}{\theta} = u_H - u_L,$$

so demand in the market with ceiling  $\theta$  evaluated at  $(\theta u_1, \theta u_2)$  equals demand in the market with ceiling 1 evaluated at  $(u_1, u_2)$ . Revenue is homogeneous of degree  $\gamma$ , since  $r(\theta u) = (\theta u)^\gamma = \theta^\gamma r(u)$ . Therefore

$$\pi_i(\theta u_i, \theta u_j; \theta) = r(\theta u_i) q_i(u_i, u_j) = \theta^\gamma \pi_i(u_i, u_j; 1).$$

The map  $d \mapsto d/\theta$  is thus a bijection between the action spaces under which every payoff in the market with ceiling  $\theta$  equals  $\theta^\gamma$  times the corresponding payoff in the market with ceiling 1.  $\square$

**Corollary 1** (Competitive escalation). *Let  $\gamma = 1$ . In the mixed strategy equilibrium, for every tolerance ceiling  $\theta$ ,*

$$\underline{s} = 0.3514 \theta, \quad \mathbb{E}[d] = 0.7029 \theta, \quad \bar{s} = 0.8557 \theta,$$

and, relative to the monopoly benchmark  $d^{\text{mon}} = \gamma\theta/(\gamma + 1)$ ,

$$\frac{\mathbb{E}[d]}{d^{\text{mon}}} = 1.406, \quad \frac{\bar{s}}{d^{\text{mon}}} = 1.711, \quad \frac{\pi^*}{\pi^{\text{mon}}} = 0.494.$$

*These ratios are independent of  $\theta$ . Average deception under duopoly therefore exceeds the monopoly level by roughly 41 per cent, and the highest level played exceeds it by roughly 71 per cent, while equilibrium profit falls to under half the monopoly benchmark.*

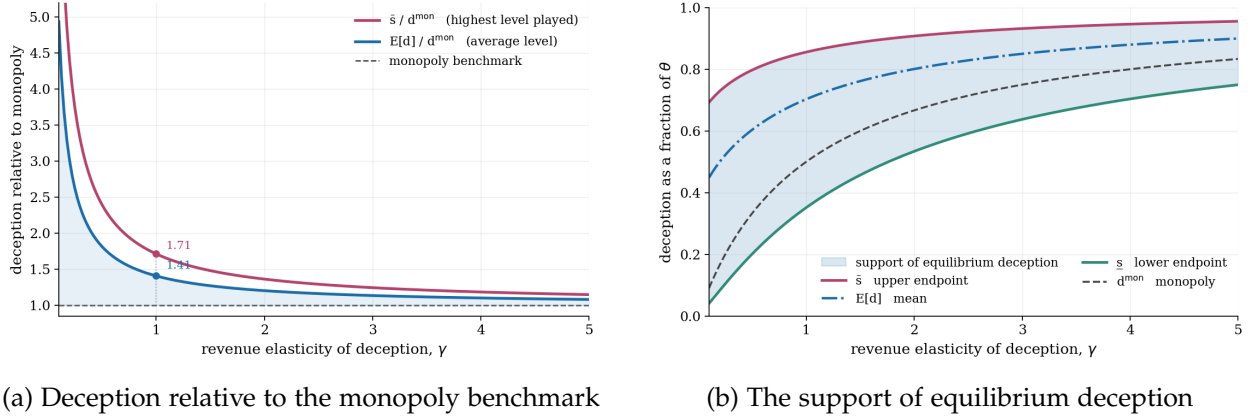
*Proof.* The solution of the normalisation condition at  $\theta = 1$  gives  $\bar{s} = 0.8557$ , and  $\underline{s} = \sqrt{\bar{s}(1 - \bar{s})} = 0.3514$ ; the mean follows from  $\mathbb{E}[d] = (\gamma + 1)\underline{s}/\gamma$ . Rescaling by Proposition 3 gives the stated levels for arbitrary  $\theta$ .  $\square$

As we sweep across various values of  $\gamma$ , the revenue elasticity of deception, we get the following results:

**Table 1.** Equilibrium deception across revenue elasticities

| $\gamma$    | Equilibrium distribution |                 |              |                           | Relative to monopoly                   |                                  |                                  |
|-------------|--------------------------|-----------------|--------------|---------------------------|----------------------------------------|----------------------------------|----------------------------------|
|             | $\underline{s}$          | $\mathbb{E}[d]$ | $\bar{s}$    | $\bar{s} - \underline{s}$ | $\frac{\mathbb{E}[d]}{d^{\text{mon}}}$ | $\frac{\bar{s}}{d^{\text{mon}}}$ | $\frac{\pi^*}{\pi^{\text{mon}}}$ |
| 0.25        | 0.104                    | 0.522           | 0.744        | 0.640                     | 2.610                                  | 3.722                            | 0.444                            |
| 0.50        | 0.201                    | 0.604           | 0.798        | 0.596                     | 1.812                                  | 2.393                            | 0.470                            |
| 0.75        | 0.283                    | 0.661           | 0.832        | 0.549                     | 1.542                                  | 1.941                            | 0.484                            |
| <b>1.00</b> | <b>0.351</b>             | <b>0.703</b>    | <b>0.856</b> | <b>0.504</b>              | <b>1.406</b>                           | <b>1.711</b>                     | <b>0.494</b>                     |
| 1.50        | 0.457                    | 0.762           | 0.887        | 0.431                     | 1.269                                  | 1.479                            | 0.506                            |
| 2.00        | 0.534                    | 0.801           | 0.908        | 0.374                     | 1.201                                  | 1.361                            | 0.513                            |
| 3.00        | 0.638                    | 0.850           | 0.932        | 0.295                     | 1.133                                  | 1.243                            | 0.522                            |
| 4.00        | 0.704                    | 0.880           | 0.946        | 0.242                     | 1.100                                  | 1.183                            | 0.527                            |
| 5.00        | 0.750                    | 0.900           | 0.955        | 0.206                     | 1.079                                  | 1.147                            | 0.530                            |





**Figure 1.** Comparative statics in the revenue elasticity  $\gamma$ .

Figure 1a plots equilibrium deception relative to the monopoly benchmark, and Figure 1b the support of the equilibrium distribution, both as fractions of  $\theta$ .

The ratio wrt monopoly falls in  $\gamma$  and diverges as  $\gamma \rightarrow 0$ . From Equation 5, the derivative of expected demand is  $D'(d) = \frac{f(d)(\theta-d)}{\theta} - \frac{1}{\theta}$ . It tells us that raising  $d$  forfeits the mass  $1/\theta$  of consumers whose tolerance is just crossed, but overtakes any rival and becoming the high firm with probability  $f(d)$ . A monopolist faces the first effect and not the second, only deceiving for the added revenue from  $\gamma$ . As  $\gamma \rightarrow 0$  the monopolist therefore stops deceiving,  $d^{\text{mon}} \rightarrow 0$ , while duopolists retain the incentive to leapfrog. Even though deception is increasing in  $\gamma$ , as the duopolists still retain some incentive to deceive while monopolists do not, the ratio  $\frac{E[d]}{d^{\text{mon}}}$  blows up as  $\gamma \rightarrow 0$ .

On the other hand, the support narrows as  $\gamma$  rises. This is because across the support, we must earn the same  $\pi^*$ . In our model, this works because the gain from increased deception perfectly cancels out the loss from lost market share on  $[\underline{s}, \bar{s}]$ . But as  $\gamma$  rises, revenue extracted increases exponentially. Thus, as  $\gamma$  rises, the revenue gains are so significant there may not be enough market share to compensate the lower end of the support, making it narrower.

## 5. Incomplete Information on Consumer Tolerance

So far, both firms knew exactly how much deception consumers would put up with. In practice, a firm rarely knows how far it can push consumers before they walk away. We model this here by making firms pick deception levels when they don't know the tolerance threshold  $\theta$ .

### 5.1. Setup

There are two states of the market. Consumer tolerance is either *fragile* or *robust*, represented by  $\theta_F$  and  $\theta_R$  respectively, with  $0 < \theta_F < \theta_R$ . Let  $\kappa \equiv \frac{\theta_R}{\theta_F} > 1$  measure how much more tolerant the robust market is. Nature draws the state, and the market is robust with probability  $\mu \in (0, 1)$ , and both firms know  $\mu$ . This could be thought of as firms learning from past experience and cases how robust such markets tend to be, but they don't know the robustness of the current market they're playing in. Firms pick  $d_1$  and  $d_2$  simultaneously without observing the realization of  $\theta$ . We assume that  $\bar{d} \geq \theta_R$ , so firms can deceive past the robust threshold if they so please. Firm  $i$  maximizes

$$\mathbb{E}_\theta[\pi_i(d_i, d_j; \theta)] = (1 - \mu)\pi_i(d_i, d_j; \theta_F) + \mu\pi_i(d_i, d_j; \theta_R)$$

## 5.2. Expected demand

Our tolerance distribution has the CDF  $G(x)_\theta = \min\left\{\frac{x}{\tilde{\theta}}, 1\right\}$ . Since the high firm serves consumers with  $\tau \geq d_H$  and the low firm serves  $d_L \leq \tau < d_H$ , demand for each firm is

$$q_L = G_\theta(d_H) - G_\theta(d_L) \quad q_H = 1 - G_\theta(d_H)$$

Let  $G_F \equiv G_{\theta_F}$  and  $G_R \equiv G_{\theta_R}$  denote the CDF in the fragile and robust states respectively. Then, expected demand can be written as

$$\mathbb{E}_\theta[q_H] = (1 - \mu)[1 - G_F(d_H)] + \mu[1 - G_R(d_H)] = 1 - G(d_H)$$

$$\mathbb{E}_\theta[q_L] = (1 - \mu)[G_F(d_H) - G_F(d_L)] + \mu[G_R(d_H) - G_R(d_L)] = G(d_H) - G(d_L)$$

where  $G$  represents the belief weighted CDF

$$G(x) \equiv (1 - \mu)G_F(x) + \mu G_R(x) \quad (7)$$

**Lemma 1** (Equivalence). *At every deception profile, a firm's expected profit in the game with uncertain tolerance ceiling equals its profit in the complete information game where the consumer tolerance has the CDF  $G$  as (7). Thus, the games have the same Nash equilibria.*

*Proof.* Since  $r(d_i) = d_i^\gamma$  regardless of whether  $\theta$  is robust or fragile, it acts as a constant when calculating expectations, so  $\mathbb{E}_\theta[\pi_i] = r(d_i)\mathbb{E}_\theta[q_i]$ . The above expressions show us that  $\mathbb{E}_\theta[q_i]$  under incomplete information equals  $q_i$  in a complete information game when the consumer has the CDF  $G(x)$ . Thus,  $\mathbb{E}_\theta[\pi_i] = \pi_i$  when  $d_1 \neq d_2$ . In a tie, each firm sells  $\frac{1 - G_\theta(d)}{2}$ , which in expectation is  $\frac{1 - G(d)}{2}$ , the same as a complete information game with CDF  $G(x)$ , giving  $\mathbb{E}_\theta[\pi_i] = \pi_i$  again. Therefore, payoffs are always equal across the two games, thus we get the same equilibria.  $\square$

Computing  $G$  piecewise, we get

$$G(x) = \begin{cases} x/\tilde{\theta}, & 0 \leq x \leq \theta_F \\ (1 - \mu) + \mu(x/\theta_R), & \theta_F < x \leq \theta_R \\ 1, & x > \theta_R \end{cases} \quad \text{where } \frac{1}{\tilde{\theta}} \equiv \frac{1 - \mu}{\theta_F} + \frac{\mu}{\theta_R} \quad (8)$$

Below  $\theta_F$ , there are two possibilities of the market tolerance, and firms can't tell what state they're in. Thus, they act off an *effective ceiling* of the probability weighted tolerance thresholds. We denote this by  $\tilde{\theta}$ , which happens to be the probability weighted harmonic mean of  $\theta_F$  and  $\theta_R$ . Since  $1/\tilde{\theta}$  is a weighted average of  $1/\theta_R$  and  $1/\theta_F$ , it lies in between them, giving us

$$\theta_F < \tilde{\theta} < \theta_R \quad (9)$$

Above  $\theta_F$ , every fragile market consumer has been lost, so the effective ceiling drops from  $1/\tilde{\theta}$  to just  $\mu/\theta_R$  as only robust consumers are left to lose. A useful property of  $G$  is that if we multiply  $\theta_F$  and  $\theta_R$  by  $t > 0$ , the new CDF  $G^t$  satisfies  $G^t(tx) = G(x)$ , giving us the same scale invariance result as Proposition 3. This allows us to normalize  $\theta_F = 1$  so the equilibrium depends only on  $\kappa$ ,  $\mu$  and  $\gamma$ .

### 5.3. The cautious equilibrium

If both firms stay at or below  $\theta_F$ , then  $G(x) = x/\tilde{\theta}$ . In this case, the game collapses to the complete information game where the known  $\theta$  equals  $\tilde{\theta}$ . Let  $F_{\tilde{\theta}}$  denote the equilibrium of Section 4.4 for a ceiling  $\theta$ , with support  $[\underline{s}, \bar{s}]$  and profit  $\pi^*(\theta)$ .

**Proposition 4** (Cautious equilibrium). *Suppose that:*

- (i)  $\bar{s} \leq \theta_F$ , and
- (ii)  $\mu \left( r(d) \frac{\theta_R - d}{\theta_R} \right) \leq \pi^*(\tilde{\theta})$  for all  $d \in (\theta_F, \theta_R]$

*Then both firms drawing from  $F_{\tilde{\theta}}$  is a mixed strategy Nash equilibrium. Each firm earns  $\pi^*(\tilde{\theta}) = \tilde{\theta}^\gamma \pi^*(1)$ , and every deception level is the complete information level with ceiling  $\theta = 1$  scaled up by  $\tilde{\theta}$ .*

*Proof.* There are three possible regions in which a firm can deviate. We can compute payoffs in each region using  $G$  by using Lemma 1.

*Case 1:*  $d \leq \theta_F$ . We have  $G = x/\tilde{\theta}$  and the result that each firm would earn the same as in the complete information game with the ceiling  $\tilde{\theta}$ .  $F_{\tilde{\theta}}$  is an equilibrium of this game, and never exceeds  $\pi^*(\tilde{\theta})$ , with equality on  $[\underline{s}, \bar{s}]$ .

*Case 2:*  $\theta_F < d \leq \theta_R$ . Since it deviates to be the high firm, they sell  $1 - G(d) = \mu \left( \frac{\theta_R - d}{\theta_R} \right)$ , which never exceeds  $\pi^*(\tilde{\theta})$  by the second assumption.

*Case 3:*  $d > \theta_R$ . The firm exceeds everyone's tolerance thresholds, earning 0.

Thus, no deviation is profitable, so  $F_{\tilde{\theta}}$  is a mixed strategy Nash equilibrium if the assumptions hold. The scaling results follow from Proposition 3.  $\square$

The first condition states that the upper end of the support never exceeds  $\theta_F$ , so the cautious equilibrium never deceives past what the fragile market would tolerate. The second tells us that it is never profitable to jump past  $\theta_F$ , despite the firm being the sole player in the region. For  $r(d) = d^\gamma$ , the monopoly profit is maximized at  $d^{\text{mon}}(\theta_R) = \frac{\gamma}{\gamma+1} \theta_R$  (Section 3.1). Thus, the second condition can be rewritten as  $\mu \pi^{\text{mon}}(\theta_R) \leq \pi^*(\tilde{\theta})$ , which tells us that gambling on the market being robust must pay less than staying cautious.

### 5.4. Uncertainty restrains deception

We compare the cautious equilibrium with two benchmarks: an average market in which the firms know the ceiling is  $\mathbb{E}[\theta] = (1 - \mu)\theta_F + \mu\theta_R$ , and full information, in which firms know  $\theta$ .

**Proposition 5.** *Under the assumptions of Proposition 4, we observe that:*

- (i) *Every deception level is strictly lower than in the average market, and expected deception is strictly lower than under full information.*
- (ii) *Firms deceive more than if they knew the market was fragile, and less than if they knew it was robust.*
- (iii) *Expected profit is strictly lower than under full information.*

*Proof.* (i) Consider  $\frac{\mathbb{E}[\theta]}{\tilde{\theta}} = [(1 - \mu)\theta_F + \mu\theta_R] \left[ \frac{1 - \mu}{\theta_F} + \frac{\mu}{\theta_R} \right] = (1 - \mu)^2 + \mu^2 + \mu(1 - \mu) \left( \kappa + \frac{1}{\kappa} \right)$ .

As  $1 = ((1 - \mu) + \mu)^2 = (1 - \mu)^2 + \mu^2 + \mu(1 - \mu)$ ,

$$\frac{\mathbb{E}[\theta]}{\tilde{\theta}} - 1 = \mu(1 - \mu) \left( \kappa + \frac{1}{\kappa} - 2 \right) = \mu(1 - \mu) \frac{(\kappa - 1)^2}{\kappa} > 0$$

Thus,  $\tilde{\theta} < \mathbb{E}[\theta]$ . We also know that levels at ceiling 1 are strictly positive as we need  $\underline{s}(1) > 0$  to ensure a strictly positive revenue and profit. Hence, by Proposition 3 and 4, the mean and support endpoints are strictly lower, since each equals  $\tilde{\theta}$  or  $\mathbb{E}[\theta]$  times the corresponding ceiling at 1. A more general argument shows that since  $d/\tilde{\theta} > d/\mathbb{E}[\theta]$ ,

$$F_{\tilde{\theta}}(d) = F_1 \left( \frac{d}{\tilde{\theta}} \right) \geq F_1 \left( \frac{d}{\mathbb{E}[\theta]} \right) = F_{\mathbb{E}[\theta]}(d)$$

Thus, for every  $d$ , firms are at least as likely to deceive by at most  $d$ , so the whole distribution shifts down. Under full information, the expected deception is  $(1 - \mu)\theta_F \mathbb{E}_1[d] + \mu\theta_R \mathbb{E}_1[d] = \mathbb{E}[\theta] \mathbb{E}_1[d]$ , which is greater than  $\tilde{\theta} \mathbb{E}_1[d]$ .

(ii) The same argument with  $\theta_F < \tilde{\theta} < \theta_R$  from (9) gives us  $F_{\theta_R} \leq F_{\tilde{\theta}} \leq F_{\theta_F}$  for all  $d$ .

(iii) Using the weighted AM-GM,  $(1 - \mu)a + \mu b \geq a^{1-\mu}b^\mu$ ,  $a, b > 0$ . If  $a = 1/\theta_F$  and  $b = 1/\theta_R$ , then  $\tilde{\theta}^{-1} > \theta_F^{-(1-\mu)}\theta_R^{-\mu}$ , so  $\tilde{\theta} < \theta_F^{1-\mu}\theta_R^\mu$ . Raising both sides to the power  $\gamma$  and using weighted AM-GM again, we get

$$\tilde{\theta}^\gamma < (\theta_F^\gamma)^{1-\mu}(\theta_R^\gamma)^\mu < (1 - \mu)\theta_F^\gamma + \mu\theta_R^\gamma$$

Multiplying the ends of the inequality by  $\pi^*(1)$ , we get  $\pi^*(\tilde{\theta}) < (1 - \mu)\pi^*(\theta_F) + \mu\pi^*(\theta_R)$ . Thus, expected profit under uncertainty (which is always  $\pi^*(\tilde{\theta})$  by Proposition 4) is lower than expected profit under full information.  $\square$

## 5.5. Overreach and busts

Now let's look at when firms start to gamble in equilibria, i.e. they play  $d > \theta_F$  with a positive probability. We say a firm *overreaches* if  $d > \theta_F$ , and a firm that overreaches *busts* if the market turns out to be fragile and thus, a firm that busts earns nothing. Say a firm overreaches with probability  $p \equiv P(d > \theta_F)$ . Since firms draw independently, the probability that neither overreaches is  $(1 - p)^2$ , so at least one overreaches with probability  $1 - (1 - p)^2$ . Thus, the probability at least one firm busts is

$$P(\text{bust}) = (1 - \mu)[1 - (1 - p)^2] \quad (10)$$

Repeating the steps from Section 4 with  $G$  instead of the uniform CDF, we find the density of the support is

$$f(d) = \frac{1}{1 - G(d)} \left[ g(d) - \frac{r'(d)\pi^*}{r(d)^2} \right]$$

with the endpoints satisfying

$$\pi^* = \frac{g(\underline{s})r(\underline{s})^2}{r'(\underline{s})}, \quad \pi^* = r(\bar{s})[1 - G(\bar{s})]$$

We can solve this numerically to check that no  $d \in (0, \theta_R)$  earns more than  $\pi^*$  on a grid. Besides the already established cautious equilibrium, we find three new equilibrium types:

- (i) *Overreach*: The support is a single interval  $[\underline{s}, \bar{s}]$  that crosses  $\theta_F$ .
- (ii) *Split*: The support is split in two, with  $[\underline{s}, s_1] \cup [s_2, \bar{s}]$ , separated around  $\theta_F$ .
- (iii) *All in*: The whole support lies above  $\theta_F$ .

Why does a split equilibrium have a gap around  $\theta_F$  while the overreach support runs right through it? Raising deception always comes with a cost and a benefit. The cost is that you lose consumers, which happens at an expected loss rate  $g(d)$ . The benefit is the gain in revenue, which is  $\frac{r'(d)}{r(d)} \cdot D(d)$ , where  $D(d)$  is expected demand from (4). Since  $D(d) = \pi^*/r(d)$ , the gain in revenue is  $\frac{r'(d)}{r(d)^2} \pi^* = \gamma \pi^*/d^{\gamma+1}$ . Therefore, firms play  $d$  only if  $g(d) \geq \gamma \pi^*/d^{\gamma+1}$ . The gain falls smoothly as  $d$  rises, but the cost is piecewise as

$$g(d) = \begin{cases} 1/\tilde{\theta}, & d < \theta_F \\ \mu/\theta_R, & d \geq \theta_F \end{cases}$$

The cost is kinked at  $\theta_F$ , causing a sudden discontinuity in the firm's choice. Whether a firm plays  $\theta_F$  in the support is determined by the sign of  $\frac{\gamma \pi^*}{\theta_F^{\gamma+1}} - \frac{\mu}{\theta_R}$ , i.e. the revenue gain at  $\theta_F$  minus the loss rate there. If it is negative, then firms play  $\theta_F$ , giving us the overreach equilibrium. If it is positive, then there's more profit to be made above  $\theta_F$ , so it is not part of the support, giving us a split equilibrium.

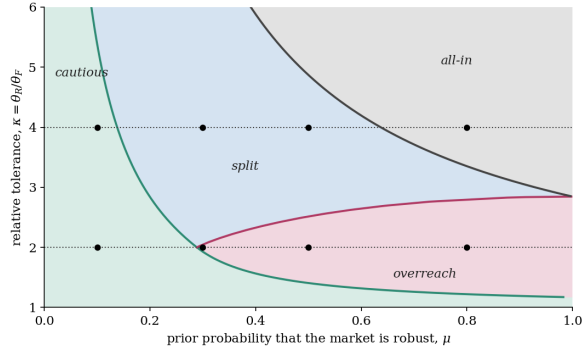
**Table 2.** Equilibrium under uncertainty ( $\gamma = 1$ ,  $\theta_F = 1$ )

| $\kappa$ | $\mu$ | Shape     | Support                              | $\mathbb{E}[d]$ | $\Pr(d > \theta_F)$ | $\Pr(\text{bust})$ |
|----------|-------|-----------|--------------------------------------|-----------------|---------------------|--------------------|
| 2        | 0.1   | cautious  | [0.370, 0.901]                       | 0.740           | 0                   | 0                  |
| 2        | 0.3   | overreach | [0.414, 1.171]                       | 0.831           | 0.033               | 0.045              |
| 2        | 0.5   | overreach | [0.481, 1.553]                       | 1.069           | 0.466               | 0.357              |
| 2        | 0.8   | overreach | [0.600, 1.678]                       | 1.314           | 0.801               | 0.192              |
| 4        | 0.1   | cautious  | [0.380, 0.925]                       | 0.760           | 0                   | 0                  |
| 4        | 0.3   | split     | [0.504, 0.949] $\cup$ [1.621, 3.171] | 1.877           | 0.573               | 0.572              |
| 4        | 0.5   | split     | [0.651, 0.923] $\cup$ [1.455, 3.372] | 2.580           | 0.896               | 0.495              |
| 4        | 0.8   | all-in    | [1.406, 3.423]                       | 2.811           | 1                   | 0.200              |

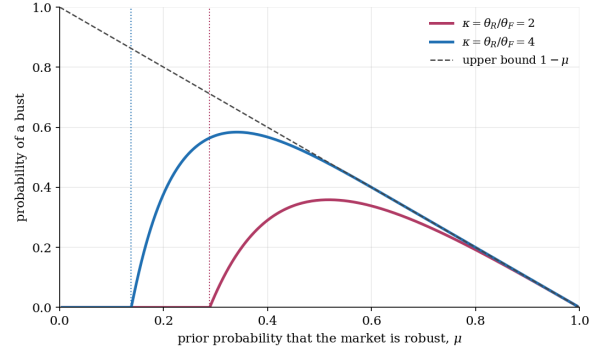
*Busts.* The risk of a bust is hump-shaped in  $\mu$  (Figure 2b). It is zero up to the cautious threshold, below which firms never overreach, and it vanishes as  $\mu \rightarrow 1$ , because a fragile market then almost never occurs. In between, firms gamble and busts are common. Risk peaks at  $\mu \approx 0.52$ , where it reaches 0.36, when  $\kappa = 2$ , and at  $\mu \approx 0.34$ , where it reaches 0.58, when  $\kappa = 4$ .

## 5.6. Summary

A firm rarely knows how much deception its customers will notice, and this uncertainty pulls it in two opposite directions. When firms think it unlikely that their customers are easy to fool, uncertainty makes them careful. A firm loses customers fastest if they turn out to be sharp, so it behaves as if its customers were sharper than they are on average. It deceives less than it would if it knew who it was selling to, and it earns less too.



(a) Equilibrium shape; dots mark the rows of Table 2



(b) Probability of a bust; dotted lines mark the end of the cautious region

**Figure 2.** Equilibrium under uncertainty across beliefs  $\mu$  and relative tolerance  $\kappa = \theta_R/\theta_F$  ( $\gamma = 1$ ,  $\theta_F = 1$ ).

Once firms think it likely enough that their customers are easy to fool, they gamble. Each sometimes pushes deception past what sharp customers would let pass, and if the customers turn out to be sharp after all, every one of them sees through the firm and it sells nothing. Such busts are not mistakes. They are the price of a bet that pays on average, and they happen most often when firms are fairly, but not fully, sure that their customers will not notice. And when easy customers would let a great deal pass, a firm that gambles gambles big, either staying safely within what sharp customers would let pass or going far beyond it.

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